

Online Project Topic Availability Checker System

How to run the project: local (XAMPP / one-click) and live (website link) — step by step

1. Quick information

ITEM	DETAILS
Technology	HTML5, CSS3, JavaScript, PHP, MySQL, XAMPP, Visual Studio Code
Source code (GitHub)	github.com/Nikilsuthar/online-project-topic-checker
Live demo link	nikilsuthar.github.io/online-project-topic-checker
Admin login	Username <code>admin</code> Password <code>admin123</code>
Student login	Username <code>student</code> Password <code>student123</code> (new students can also register)

2. Four ways to run — which one to use?

METHOD	TYPE	WHAT YOU NEED	BEST FOR
A. XAMPP Recommended	Local	XAMPP installed. PHP + real MySQL database, exactly as written in the synopsis.	College submission, viva, showing phpMyAdmin database
B. START.bat	Local	Nothing to install. Double-click one file (uses built-in PHP + SQLite file database).	Backup if XAMPP is not installed or not working
C. GitHub Pages link	Live	Only a browser. Open the link on any phone or laptop.	Quick demo, sharing a link
D. Render.com	Live	Free Render account (login with GitHub). Runs the real PHP code online.	A live website of the full PHP version

What is the difference? **Local** means it runs on your own computer at `http://localhost/...` — only that computer can open it. **Live** means it runs on the internet with a link anyone can open.

3. Method A — Run with XAMPP (PHP + MySQL)

Step 1: Install XAMPP (only once)

1. Download XAMPP for Windows from <https://www.apachefriends.org> and install it (keep default options, install to `C:\xampp`).

Step 2: Put the project in htdocs

1. Get the project folder `Online_Project_Topic_Checker` (from the ZIP file you were given, or from GitHub: open the GitHub link → green **Code** button → **Download ZIP** → extract).
2. Copy the folder into `C:\xampp\htdocs\`
Result: `C:\xampp\htdocs\Online_Project_Topic_Checker\index.php`

If you downloaded from GitHub, the folder is named `online-project-topic-checker-main`. Rename it to `Online_Project_Topic_Checker` (or use that name in the address in Step 4).

Step 3: Start the servers

1. Open **XAMPP Control Panel**.
2. Click **Start** next to **Apache** and next to **MySQL**. Both should turn green.

Step 4: Open the website

1. Open a browser and go to: http://localhost/Online_Project_Topic_Checker/
2. Login with `admin / admin123` or `student / student123`.

SQL / database setup — automatic

You do **not** need to create the database by hand. The first time the website opens, it automatically:

- creates the MySQL database `project_topic_checker`,
- creates the tables `users`, `topics` and `topic_requests`,
- adds sample topics and the demo accounts.

To see the database (good for the viva): open <http://localhost/phpmyadmin> → click `project_topic_checker` on the left → open any table.

Optional manual way: in phpMyAdmin click **Import** → choose `database.sql` from the project folder → **Go**. (Only creates empty tables; the website adds the sample data.)

Database settings are in `config.php` (XAMPP default: user `root`, empty password). Change them only if your MySQL has a password.

Common problems

PROBLEM	SOLUTION
Apache does not start (port 80 busy)	Close Skype / other web servers, or in XAMPP click Config → httpd.conf , change <code>Listen 80</code> to <code>Listen 8081</code> , restart, and open <code>http://localhost:8081/Online_Project_Topic_Checker/</code>
MySQL does not start (port 3306 busy)	Another MySQL is running. Stop it from Windows Services, or restart the PC and start XAMPP first.
Page shows "Database connection failed"	MySQL is not started. Start MySQL in XAMPP and refresh.
"Not Found" error	Folder name in the address does not match the folder in htdocs. Check spelling.
Want to start with fresh data	phpMyAdmin → select <code>project_topic_checker</code> → Operations → Drop the database . Open the website again and it is recreated.

4. Method B — One-click run with START.bat (no installation)

1. Use the project folder from the **ZIP file** (it contains the `_tools` folder with PHP). *The GitHub download does not include `_tools`, so START.bat only works from the ZIP.*
2. Double-click **START.bat**. A black window opens and the browser opens **`http://localhost:8080`**.
3. Keep the black window open while using the website. Close it to stop.

This method stores data in a small file database (SQLite) in the `data` folder instead of MySQL. All features work the same. To reset, close the window and delete the `data` folder.

5. Method C — Live demo link (GitHub Pages)

1. Open **`https://nikilsuthar.github.io/online-project-topic-checker/`** on any phone or computer.
2. Tap **Use** next to a demo account, then **Login**.

GitHub Pages can only host HTML/JavaScript, not PHP or MySQL. So this link runs a browser version of the same system (same screens, rules and checks). Data is saved in that browser only; use **Reset demo data** to start fresh. For PHP + MySQL use Method A or D.

6. Method D — Live PHP website on Render.com (free)

The GitHub repository already contains the deploy files (`Dockerfile` and `render.yaml`). You only need to connect it:

1. Go to **`https://render.com`** → **Get Started** → sign up with **GitHub** (account *Nikilsuthar*).
2. In the Render dashboard click **New +** → **Blueprint**.
3. Connect / select the repository **online-project-topic-checker**.
4. Render reads `render.yaml` and shows the service **project-topic-checker** (Free plan). Click **Apply / Deploy Blueprint**.
5. Wait about 5–10 minutes for the first build. When it says **Live**, open the link shown at the top, for example **`https://project-topic-checker.onrender.com`** (the exact name may differ).

Free plan notes: the site sleeps after ~15 minutes without visitors; the first open after that takes about 1 minute. Open the link a few minutes before presenting. By default the live site uses a built-in SQLite database, so data goes back to the sample data when the service restarts.

Optional: use a real MySQL database online

1. Create a free cloud MySQL database (for example on **aiven.io** or another MySQL host) and note its host, port, database name, user and password.
2. In Render open the service → **Environment** → add: `PTC_DB_DRIVER = mysql`, `DB_HOST`, `DB_PORT`, `DB_NAME`, `DB_USER`, `DB_PASS`, and `DB_SSL = 1` if the provider requires SSL.
3. Save — Render redeploys, and the tables are created automatically on first open.

7. Updating the code on GitHub

After changing code in VS Code, run these in the project folder (Render updates the live site automatically):

```
git add -A → git commit -m "your message" → git push
```

8. Suggested demo flow for the presentation

1. **Student login** → Dashboard shows total, available and selected topics.
2. **Topic Checker** → type *PG Finder Website* → "Topic Appears to be Available" → add a description → **Submit Topic for Approval**.
3. Type *Library Management* → "Topic Not Available" (already selected — duplicate prevented).
4. Type *Online Exam Result System* → "Similar Topics Found" with % match.
5. Type *Online Management System* → validation: topic is too generic.
6. **Logout** → **Admin login** → Topic Requests → **Approve** (try Reject without a remark to show validation).
7. **Manage Topics** → add a topic, try adding a duplicate (blocked), edit / release / delete.
8. **Reports** → summary, department-wise report, selected/available lists, print and CSV download.
9. (XAMPP) Show the tables in **phpMyAdmin**.

9. Project structure

FILE / FOLDER	PURPOSE
index.php , register.php , logout.php	Login, student registration, logout
config.php	Database settings
includes/	Database connection + automatic setup, common functions, validation, topic matching, page layout
student/	Dashboard, Topic Checker, Project Topics, Topic Requests
admin/	Dashboard, Manage Topics, Topic Requests, Students, Reports
assets/	CSS design and JavaScript (form validation, live search/filter)
database.sql	MySQL table structure (optional import)
demo/index.html	Browser version used for the live demo link
START.bat , _tools/	One-click local run without XAMPP
Dockerfile , render.yaml	Live hosting on Render.com